Thermochimica Acta 566 (2013) 44–49

Contents lists available at SciVerse ScienceDirect

Thermochimica Acta

journal homepage: www.elsevier.com/locate/tca

Thermodynamic assessment of the Bi–Er and the Bi–Dy systems

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ARTICLE INFO

Article history:
Received 4 March 2013
Received in revised form 10 May 2013
Accepted 17 May 2013
Available online 25 May 2013

Keywords:
CALPHAD
Phase diagram
First-principles calculations
The Bi–Er system
The Bi–Dy system

ABSTRACT

Based on the available experimental data, the Bi–Er and the Bi–Dy binary systems have been assessed thermodynamically using the CALPHAD (CALculaton of PHAse Diagram) technique. The formation enthalpies at 0 K of the compounds, Bi_3Er_5, BiEr, Bi_3Dy_5 and BiDy, were calculated by the first-principles method and Birch–Murnaghan equation of state, and the calculated results were used in the present thermodynamic optimization. All these compounds in the Bi–Er and the Bi–Dy systems were treated as stoichiometric compounds. The liquid phase was modeled as a substitutional solution phase based on random mixing of the constituent atoms and the excess Gibbs energy was formulated with the Redlich–Kister polynomial. Good agreements were obtained between the calculated results and the experimental data for both the thermochemical properties and the phase equilibrium relations.

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1. Introduction

In order to increase the phonon-scattering in the Bi_2Te_3-based thermoelectric materials, doping rare earth (RE) elements is an effective method, resulting in raising the Seeback coefficient and improving thermoelectric properties [1–5]. Consequently the Bi–RE systems deserve to be studied so as to accomplish the Bi–RE–Te thermodynamic database for designing high-performance thermoelectric materials. In order to better understand the interactions of Er and Dy with Bi and design high performance thermoelectric materials with RE elements, it is important to study the thermochemical properties and the phase equilibrium relations concerning the Bi–Er and the Bi–Dy systems.

As part of the Bi–RE–Te thermodynamic database, this work presents the thermodynamic descriptions of the Bi–Er and the Bi–Dy binary systems. Especially, before these two systems are thermodynamically optimized, the necessary thermodynamic properties of the Bi–Er and the Bi–Dy compounds are calculated through the first-principles method. And the calculated results are used in the CALPHAD modeling together with the available experimental data.

2. Evaluation of the available information

2.1. The Bi–Er system

The phase equilibria of the Bi–Er binary system were previously measured by Yoshihara et al. [6]. It was reported that two intermetallic compounds, Bi_3Er_5 and BiEr, exist in this system. Later, by means of differential thermal analysis (DTA), X-ray diffraction phase analysis (XRD) and microstructural analysis, Abulkhaev [7] reinvestigated the phase diagram of the Bi–Er system. Based on the investigated results, the Bi_3Er_5 phase is formed from a peritectic reaction, $L + BiEr \leftrightarrow Bi_3Er_5$ at 1700 K, whereas the BiEr phase melts congruently at about 2080 K. There are two eutectic reactions, $L \leftrightarrow$ Hcp_A3 (Er) + Bi_3Er_5 at 1434 K and $L \leftrightarrow$ Rhomb (Bi) + BiEr at 542 K. In view of the fact that the solubility of each terminal phase is very limited, they are treated as pure element phases in the present work. The details of the invariant reactions in this system are shown in Table 1.

In addition, the thermochemical data of the Bi–Er binary system were reported by several groups. Colinet et al. [8] and Takeuchi and Inoue [9] calculated the value of the enthalpy of mixing of the liquid phase at 50 at.% Bi by Miedema model. The accuracy of calculated results depended on the parameters used in Miedema model which were modified regularly. Therefore, the calculate results from Ref. [9] were newer and more reasonable, and a higher weight was given to them than to those from Ref. [8]. The formation enthalpy of the intermetallic compound BiEr was measured in Ref. [10] by direct calorimetry method and in Refs.

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http://dx.doi.org/10.1016/j.tca.2013.05.024

Table 1
Invariant reactions of the Bi–Er binary system.

Invariant reaction	Reaction type	T (K)	x^L_{Bi}	Reference
L ↔ Bi_3Er_5 + Hcp_A3 (Er)	Eutectic	1434	0.2	[7]
		1434	0.19	This work
L + BiEr ↔ Bi_3Er_5	Peritectic	1700	~0.325	[7]
		1700	0.315	This work
L ↔ BiEr	Congruent	2080	0.5	[7]
		2080	0.5	This work
L ↔ BiEr + Rhomb_A7 (Bi)	Eutectic	542	~0.99	[7]
		541	0.99	This work

Table 2
Invariant reactions of the Bi–Dy binary system.

Invariant reaction	Reaction type	T (K)	x^L_{Bi}	Reference
L ↔ Bi_3Dy_5 + Hcp_A3 (Dy)	Eutectic	1420	~0.185	[14]
		1420	0.175	This work
L + BiDy ↔ Bi_3Dy_5	Peritectic	1719	~0.335	[14]
		1719	0.33	This work
L ↔ BiDy	Congruent	2048	0.5	[14]
		2048	0.5	This work
L ↔ BiDy + Rhomb_A7 (Bi)	Eutectic	542	~0.99	[14]
		543.5	0.995	This work

[11,12] by EMF method. The experiment data of limiting values of the formation enthalpy of the Bi_3Er_5 phase were also given in Refs. [11,12]. Those of both intermetallic compounds BiEr and Bi_3Er_5 were predicted in Ref. [8] by Miedema model.

2.2. The Bi–Dy system

The paritial phase diagram of the Bi–Dy binary system was previously measured by Gambino [13], with the composition range covering 0–60 at.%. It was reported that two intermetallics, Bi_3Dy_5 and BiDy, and three terminal solution phases, Hcp_A3 (Dy), Bcc_A2 (Dy) and Rhomb_A7 (Bi), existed in this system. Later, by using DTA, XRD and microstructure analysis, Abulkhaev [14] remeausured the phase diagram of the Bi–Dy binary system. According to measured data, the compound Bi_3Dy_5 is formed by the peritectic reactions, L + BiDy ↔ Bi_3Dy_5 at 1693 K, and the compound BiDy melts congruently at about 2048 K. There are two eutectic reactions, L ↔ Hcp_A3 (Dy) + Bi_3Dy_5 at 1428 K and L ↔ Rhomb (Bi) + BiDy at 544 K. Because the terminal phases in both sides have very narrow homogeneity ranges, they were treated as pure element phases in the present assessment. The details of the invariant reactions in this system are shown in Table 2.

Moreover, several literatures reported the thermochemical data of the Bi–Dy binary system. Using of Miedema model, the enthalpy of mixing of the liquid phase at 50 at.% Bi was calculated in Refs. [8,9]. According to the same reason as for the Bi–Er system, the higher weight was given to the data from Ref. [9] in present assessment. Borsese et al. [15] measured the values of the formation enthalpies of the intermetallic compounds Bi_3Dy_5 and BiDy by calorimetry method and Colinet et al. [8] predicted these two values by Miedema model.

3. Calculation method

3.1. First-principles calculations

First-principles calculations have been demonstrated to be a powerful method to assist CALPHAD modeling. In the present work, in order to supplement the valid data of thermochemical properties for intermetallics and have more reliable descriptions of the thermodynamic parameters for systems, the first-principles method was employed to calculate the formation enthalpies of the binary compounds in the Bi–Er and the Bi–Dy systems. These first-principles calculation results were considered to be reasonable and were used in the CALPHAD optimizing.

The scalar relativistic all-electron Blöchl's projector augmented-wave (PAW) [16,17] method within the generalized gradient approximation (GGA) was adopted, as implemented in the highly-efficient Vienna ab initio simulation package (VASP) [18,19]. For the GGA exchange-correlation functional, the Perdew–Burke–Ernzerhof parameterization (PBE) [20–22] was employed. A plane-wave energy cutoff of 400 eV was held constant for the A_xB_y compounds (A = Bi, B = Er or Dy, and x and y are the stoichiometric numbers of the elements A and B respectively in the related compound), and Brillouin zone integrations were performed using Monkhorst–Pack [23,24] K-point meshes, and the K-point meshes of compounds were $12 \times 12 \times 12$. The total energy was converged numerically to less than 1×10^{-6} eV/unit with respect to electronic, ionic and unit cell degrees of freedom. All calculations were performed using the "high" setting within VASP to avoid wrap-around errors. Furthermore, the energy vs. volume equations of state (EOS) [25,26] was used to calculate the minimum of the total energy and lattice constants.

In order to calculate the minimum of total energy of each compound, the volume-dependent energy curves obtained from first-principles calculations are fitted by means of the third-order Birch–Murnaghan equation of state [27–29] given by:

$$E(V) = E_0 + \frac{9V_0B_0}{16}\left\{\left[\left(\frac{V_0}{V}\right)^{2/3} - 1\right]^3 B_0' + \left[\left(\frac{V_0}{V}\right)^{2/3} - 1\right]^2 \left[6 - 4\left(\frac{V_0}{V}\right)^{2/3}\right]\right\} \quad (1)$$

where V is the volume of primitive cell; $E(V)$ is the total energy corresponding to V. The equilibrium properties fitted with EOS include

the equilibrium volume (V_0), equilibrium energy (E_0), equilibrium modulus (B_0) and its pressure derivative (B_0').

In addition, the formation enthalpies of the intermetallic compounds were calculated by the following equation:

$$\Delta H\left(A_xB_y\right) = E_{\text{total}}\left(A_xB_y\right) - xE_{\text{total}}(A) - yE_{\text{total}}(B) \quad (2)$$

where $E_{\text{total}}\left(A_xB_y\right)$, $E_{\text{total}}(A)$ and $E_{\text{total}}(B)$ are the calculated total energies (per atom at $T = 0$ K) of the intermetallic compounds, pure A and B, respectively.

3.2. Thermodynamic modeling

The Gibbs energy functions of pure elements were taken from the SGTE (Scientific Group Thermodata Europe) database [30]. Different models were employed to describe the liquid phase and the intermetallic compounds in the Bi–Er and the Bi–Dy binary systems.

3.2.1. Liquid phase

The liquid phase was modeled as a substitutional solution phase based on random mixing of the constituent atoms. The Gibbs energy of the liquid phase was expressed with the following equation [31]:

$$G_m^{Liq} = x_A^0 G_A^{Liq} + x_B^0 G_B^{Liq} + RT(x_A \ln x_A + x_B \ln x_B) + x_A x_B \sum_{i=0}^{n} {}^iL_{A,B}^{Liq}(x_A - x_B)^i \quad (3)$$

where x_i is the mole fraction of the component i (i = A or B, A – B = Bi–Er or Bi–Dy), R is the gas constant, and ${}^iL_{A,B}^{Liq}$ is the interaction parameter between species A and B, which is formulated as a linear function of temperature as ${}^iL_{A,B}^{Liq} = a + bT$, and the coefficients a and b are the thermodynamic parameters to be optimized.

3.2.2. Intermetallic compounds

In the Bi–Er and the Bi–Dy binary systems, there are four intermetallic compounds totally, Bi_3Er_5, BiEr, Bi_3Dy_5 and BiDy. All these compounds were treated as stoichiometric compounds A_mB_m because the homogeneity ranges of the compounds are very narrow with no reported experimental data. Meanwhile, due to lack of the heat capacity data of these compounds, the Gibbs energy per mole of atoms was expressed according to the Neumann–Kopp rule, as follows:

$$G_m^{A_mB_n} = \frac{m}{m+n}{}^0G_A^{SER} + \frac{n}{m+n}{}^0G_B^{SER} + c + dT \quad (4)$$

where the parameters c and d are to be assessed.

4. Results and discussion

4.1. First principles calculations

In the present work, the Birch–Murnaghan equation of state (Eq. (1)) was used to determine the minimum of total energy and the lattice constants. The experimental lattice parameters were used as the starting point to perform the structural calculations for Bi, Er, Dy, Bi_3Er_5, BiEr, Bi_3Dy_5 and BiDy [6,32–35]. The total energies of these four compounds were assessed for different b/a and c/a ratio around the equilibrium cell. And these two equilibrium parameters, b/a and c/a, were fixed by calculating the equilibrium volume using Birch–Murnaghan equation of state (Eq. (1)). In Table 3 the optimized results of the lattice constants a, b and c are summarized and compared with the available experimental results taken from Refs. [6,32–35]. Notably these theoretical values calculated from the first-principles method slightly overestimate the lattice constants about 1% than experimental values for Bi, Er, Dy, Bi_3Er_5, BiEr, Bi_3Dy_5 and BiDy.

Based on the minimum of total energy by Eqs. (1) and (2), the formation enthalpies of the binary compounds in the Bi–Er and the Bi–Dy systems were calculated. The comparison between the calculated results and the experimental data is discussed following. For the Bi–Er system, the experiment data of the BiEr phase are given in Ref. [10] by direct calorimetry and in Refs. [11,12] by EMF method. The value of BiEr calculated by the first-principles method is conformable with the experimental data reported by Ref. [10], and the relative deviation is about 9%. Because the EMF method used in Refs. [11,12] has greater error than the direct calorimetry method used in Ref. [10] and because the results calculated by the first-principles method agree very well with the experiment data from Ref. [10], a higher weight is given to the data from Ref. [10] and the present first-principles calculations than to those from Refs. [11,12]. For the Bi–Dy system, the results of first-principles calculations of BiDy are consistent with the data reported by Borsese et al. [15], with the relative deviation within 10%. And for the Bi_3Dy_5 compound, the relative deviation is about 14%. It is indicated that the present first-principles calculations are reasonable and acceptable. Furthermore, all the experiment data from Ref. [15] are plotted

Table 3
The lattice constants a, b and c compared to other available experimental data [6,32–35].

Phase	Pearson symbol	Space group	a (Å)	B (Å)	C (Å)	Reference
Bi	hR2	R-3m	4.5556	–	12.0249	First-principle
			4.5446	–	11.8591	Experiment [32]
Er	hP2	$P6_3/mmc$	3.513	–	5.517	First-principle
			3.495	–	5.486	Experiment [33]
Dy	hP2	$P6_3/mmc$	3.511	–	5.603	Fist-principle
			3.4936	–	5.506	Experiment [34]
Bi_3Er_5	hP16	Pnma	8.171	9.446	11.889	First-principle
			8.09	9.349	11.806	Experiment [35]
BiEr	cF8	Fm-3m	6.2644	–	–	First-principle
			6.2023	–	–	Experiment [6]
Bi_3Dy_5	oP32	Pnma	8.235	9.516	12.011	First-principle
			8.155	9.432	11.934	Experiment [35]
BiDy	cF8	Fm-3m	6.3071	–	–	First-principle
			6.2491	–	–	Experiment [6]

Table 4
The formation enthalpies obtained by the first-principles calculations, the Miedima model [8] and reported experimental data [10–12,15].

Phase	Experimental (J/mol)	First-principles in this work at 0 K (J/mol)	Miedima model (J/mol)
Bi_3Er_5	≤−81,000 [11] ≤−77,000 [12]	−66,035	−69,000 [8]
BiEr	−90,000 [10] −109,000 [11] −103,000 [12]	−82,069	−75,000 [8]
Bi_3Dy_5	−81,600 [15]	−69,752	−73,000 [8]
BiDy	−96,200 [15]	−86,556	−79,000 [8]

Table 5
Thermodynamic parameters of the Bi–Er binary system obtained in this work.

Phase	Thermodynamic parameters
Liquid (Bi, Er)	${}^0L^{Liq}_{Bi,Er} = -190000 - 1.241T$ ${}^1L^{Liq}_{Bi,Er} = 7000 + 2.212T$ ${}^2L^{Liq}_{Bi,Er} = 42000 + 4.5T$ ${}^3L^{Liq}_{Bi,Er} = -22700 - 0.225T$
Bi_3Er_5	$G^{Bi_3Er_5}_{Bi:Er} = -70280 + 3.99T + 0.625GHSERER + 0.375GHSERBI$
BiEr	$G^{BiEr}_{Bi:Er} = -83620 + 2.9T + 0.5GHSERER + 0.5GHSERBI$

in Fig. 6, and they accord well with the assessed results. The present first-principles calculation results, the related experimental data and Miedema model predictions are listed in Table 4.

4.2. Thermodynamic assessments

On the basis of the thermodynamic data of the pure elements Er, Dy and Bi compiled by Dinsdale [30] and the experimental data of both the thermochemical properties and the phase equilibrium relations, the thermodynamic parameters of the Bi–Er and the Bi–Dy binary systems were assessed using the PARROT module of the Thermo-Calc software [36]. Tables 5 and 6 list the optimized thermodynamic parameters for the Bi–Er and the Bi–Dy systems, respectively. And the round-off policy of parameters accords with 1 J/mol at 1000 K. As discussed in the next sections, most of the thermodynamic properties and the phase boundary data are well reproduced.

4.2.1. The Bi–Er binary system

The calculated phase diagram is shown in Fig. 1 together with the experimental data from Abulkhaev [13]. It can be seen that the calculated phase diagram is in reasonable agreement with the experimental data. The calculated peritectic temperature for the formation of the intermetallic compound Bi_3Er_5 is 1700 K, the congruent melting temperature of BiEr is 2080 K, and the eutectic temperatures of the reactions, L ↔ Hcp_A3 (Er) + Bi_3Er_5 and L ↔ Rhomb (Bi) + BiEr, are 1434 and 541 K, respectively. The details of the invariant reactions in this binary system together with the data from the literatures are listed in Table 1 for comparison.

Fig. 2 shows the calculated mixing enthalpies of the liquid phase at 2500 K, together with the results reported by Refs. [8,9], from

Table 6
Thermodynamic parameters of the Bi–Dy binary system obtained in this work.

Phase	Thermodynamic parameters
Liquid (Bi, Dy)	${}^0L^{Liq}_{Bi,Dy} = -198000 + 1.619T$ ${}^1L^{Liq}_{Bi,Dy} = 11500 + 1.272T$ ${}^2L^{Liq}_{Bi,Dy} = 51500 + 4.132T$ ${}^3L^{Liq}_{Bi,Dy} = -31203 - 0.057T$
Bi_3Dy_5	$G^{Bi_3Dy_5}_{Bi:Dy} = -78840 + 8T + 0.625GHSERDY + 0.375GHSERBI$
BiDy	$G^{BiDy}_{Bi:Dy} = -92664 + 7.55T + 0.5GHSERDY + 0.5GHSERBI$

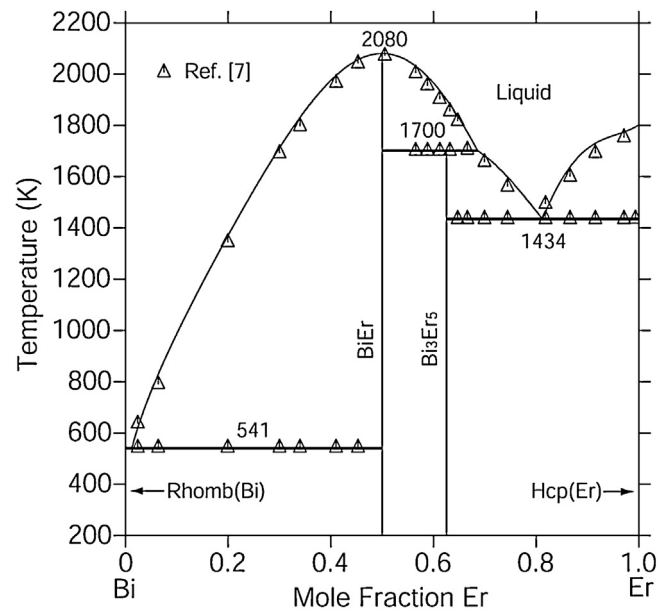

Fig. 1. The assessed Bi–Er phase diagram in comparison with the experimental data [7].

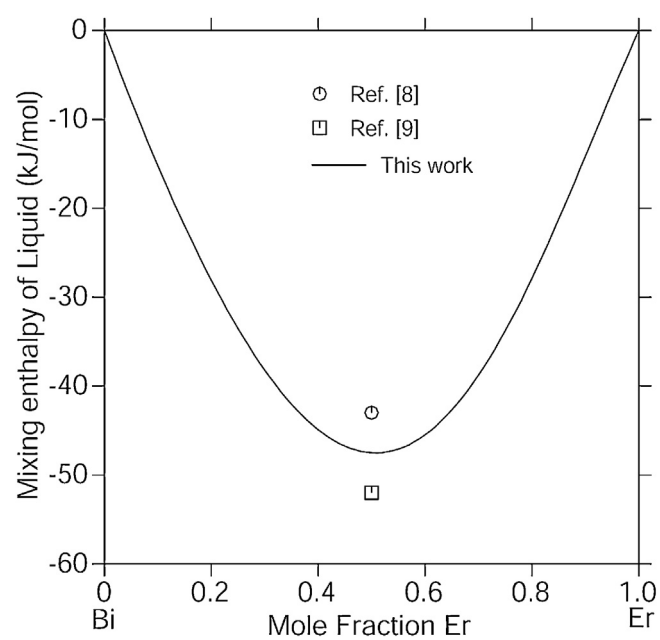

Fig. 2. The mixing enthalpy of liquid Bi–Er alloys at 2500 K: comparison between calculated values and reported data [8,9]. Reference states are liquid Er and liquid Bi.

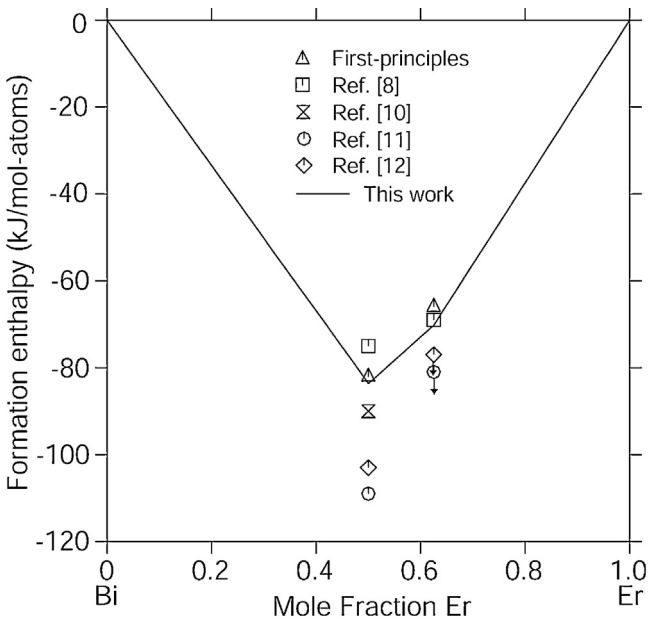

Fig. 3. The formation enthalpy of Bi–Er intermetallic compounds at 300 K: comparison between calculated values with reported data [8,10–12]. Reference states are Hcp (Er) and Rhomb (Bi).

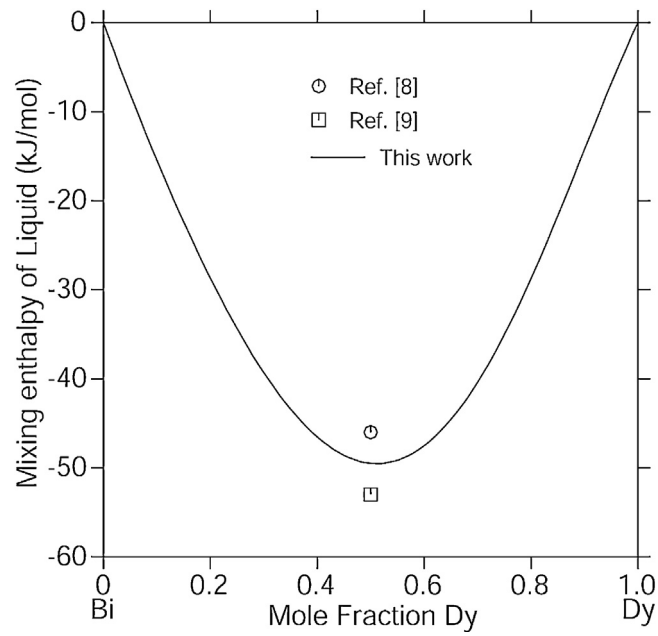

Fig. 5. The mixing enthalpy of liquid Bi–Dy alloys at 2500 K: comparison between calculated values and reported data [8,9]. Reference states are liquid Dy and liquid Bi.

which it can be seen that a good agreement is obtained. Fig. 3 represents the calculated formation enthalpies of intermetallic compounds at 300 K together with the experiment data [10] and the first-principles calculations. The calculated values agree very well with the reported data and the present first-principles calculations.

4.2.2. The Bi–Dy binary system

Fig. 4 shows the calculated phase diagram of the Bi–Dy system together with the experimental data taken from Ref. [14]. The peritectic temperature for the formation of the intermetallic compound Bi_3Dy_5 is 1719 K, and the congruent melting temperature of BiDy is 2023 K, the eutectic temperatures of the reactions, L ↔ Hcp_A3 (Dy) + Bi_3Dy_5 and L ↔ Rhomb (Bi) + BiDy, are 1420 and 543.5 K, respectively. Table 2 lists all the calculated invariant reactions in the Bi–Dy system. And the calculated results are in good agreement with most of the experimental data.

Fig. 5 presents the calculated mixing enthalpies of the liquid phase at 2500 K, together with the results reported by Refs. [8,9], from which it can be seen that a good agreement is obtained. The calculated formation enthalpies of the intermetallic compounds at 300 K are compared with the first-principles calculations, as shown in Fig. 6. The calculated values agree very well with the present first-principles calculations and the data reported by Borsese et al. [15].

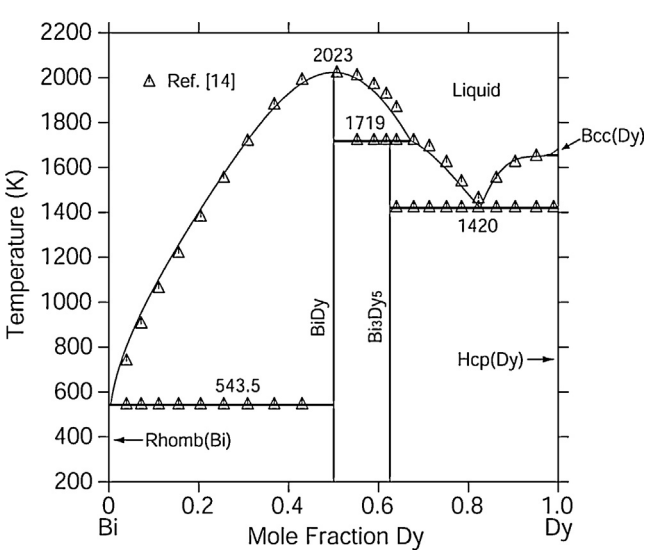

Fig. 4. The calculated Bi–Dy phase diagram in comparison with the experimental data [14].

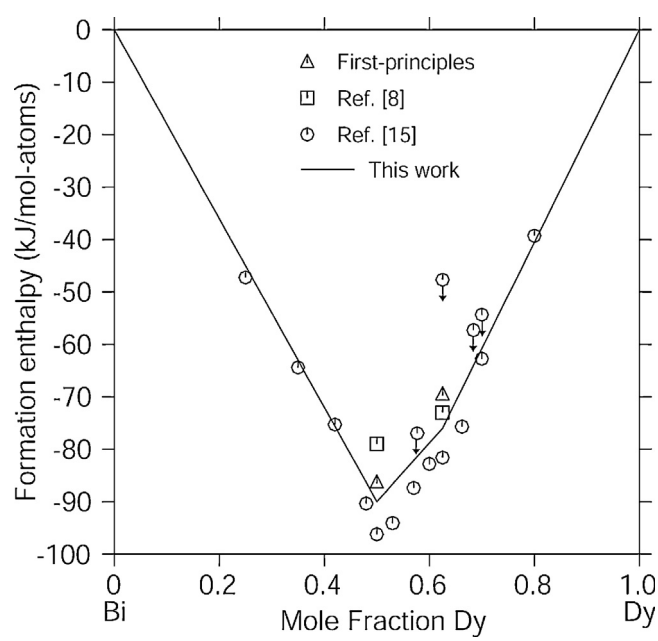

Fig. 6. The formation enthalpy of Bi–Dy intermetallic compounds at 300 K: comparison between calculated values with reported data [8,15]. Reference states are Hcp (Dy) and Rhomb (Bi).

5. Conclusions

Based on the present first-principles calculations and the experimental data of the phase diagrams and the thermodynamic properties available in literatures, the Bi–Er and the Bi–Dy binary systems were thermodynamically modeled critically. A set of self-consistent thermodynamic parameters for each of these two systems have been obtained. For both of the systems, the comprehensive comparisons show that the calculated phase diagrams and the thermodynamic properties are in good agreement with the available experimental data and the present first-principles calculations.

Acknowledgements

The authors would like to acknowledge National Natural Science Foundation of China (no. 51271027) and National High Technology Research and Development Program of China (no. 2013AA031601) for the financial supports. Thanks to Royal Institute of Technology and CompuTherm LLC for supplying Thermo-Calc and Pandat Software packages respectively.

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